

SECTION-01

1a. Digital transformation is creating new or modifying existing business processes, culture, and customer experiences to meet changing business and market requirements. Explain how digital transformation has brought revolution in retail purchases with an example.

➡ Digital transformation in retail means integrating modern technologies to enhance every aspect of the shopping experience — from inventory management to customer interaction. It's not just about selling online; it's about reimagining how retail works.

Key Changes in Retail Due to Digital Transformation

- **Omnichannel Shopping:** Customers can seamlessly switch between online and offline experiences — browsing online, picking up in-store, or returning via app.
- **Personalized Experiences:** AI and data analytics allow retailers to tailor product recommendations, promotions, and even pricing based on customer behavior.
- **Real-Time Inventory Management:** Retailers use cloud-based systems to track stock across locations, reducing shortages and overstock.
- **Automated Checkout:** Technologies like RFID, mobile payments, and self-checkout kiosks speed up transactions and reduce human error.
- **Virtual Try-Ons & AR:** Augmented reality lets customers “try” clothes, makeup, or furniture virtually before buying.

Example: Amazon Go — The Checkout-Free Store

Amazon Go is a prime example of digital transformation in action:

- **How it works:** Customers enter the store using the Amazon Go app, pick up items, and simply walk out. Sensors and computer vision track what they take, and the app charges them automatically.
- **Impact:**
 - Eliminates checkout lines
 - Reduces staffing needs
 - Enhances customer convenience
 - Collects rich data on shopping habits for better inventory and marketing decisions

This model redefines the traditional retail experience, blending physical presence with digital intelligence.

1b. Diagnostic imaging procedures are cutting-edge technology, but at the same time they are an unpleasant experience for patients - and even more for paediatric patients. Explain how design thinking helped Doug Dietz, an industrial designer, create a scanner experience that children loved.

➡ Doug Dietz, an industrial designer at GE Healthcare, had developed a state-of-the-art MRI scanner. But during a hospital visit, he witnessed a young girl crying in fear before entering the machine. He realized that while the technology was advanced, the experience was terrifying for children.

How Design Thinking Helped:

Doug applied **design thinking**, a human-centered approach to problem-solving, which involves:

- **Empathy:** He observed children in daycare and spoke to child-life specialists to understand their fears.
- **Define:** He reframed the problem — not just designing a better machine, but creating a better experience.
- **Ideate:** Collaborated with museum experts, hospital staff, and volunteers to brainstorm creative solutions.
- **Prototype & Test:** Developed themed environments around the scanners — like pirate ships, space adventures, and jungle safaris.

How Doug Dietz Made the Adventure Series MRI Scanner – Step-by-Step

1. Realization at the Hospital

Doug visited a hospital to see his newly designed MRI scanner in use. Instead of praise, he saw a young girl crying in fear. He learned that **up to 80% of children were sedated** for scans because they were too scared to stay still. This moment made him realize the machine was technically brilliant but emotionally intimidating.

2. Learning Design Thinking

Doug attended a workshop at **Stanford's d.school**, where he learned the **human-centered design process**. This gave him tools to approach the problem from the user's (child's) perspective.

3. Empathy Research

He began observing children at **daycare centers** and spoke with **child-life specialists** to understand what pediatric patients go through emotionally. He also interviewed hospital staff and parents to gather insights.

4. Building a Cross-Functional Team

Doug didn't work alone. He formed a team that included:

- GE volunteers
- Designers from a **local children's museum**
- Doctors and nurses from two hospitals Together, they brainstormed how to make the MRI experience feel like an **adventure**, not a medical procedure.

5. Creating Themed Prototypes

They developed **themed MRI rooms** like:

- Pirate ship
- Jungle safari
- Space mission These rooms included:
- Wall art and props
- Ambient sounds and lighting
- Staff trained to act out the story (e.g., “Don’t rock the boat!”)

6. Testing in Real Hospitals

They installed these themed rooms in actual hospitals and observed how children responded. The results were dramatic:

- Kids were excited to “go on an adventure”
- Sedation rates dropped significantly
- Parents and staff reported better experiences

Outcome: The Adventure Series:

- Scanners were transformed into immersive experiences.
- Children felt like they were going on an adventure, not undergoing a medical procedure.
- Sedation rates dropped dramatically, and hospitals reported improved patient satisfaction.

Doug’s story shows that **empathy and creativity** can turn fear into joy — even in the most clinical settings

2a. Identify the tasks involved in the process of receiving Admission Ticket for semester exams in your college. Which of the identified tasks can be automated and illustrate automation of one task

➡ Tasks Involved in Receiving Admission Ticket:

1. Verification of Eligibility

- Checking attendance, internal marks, and fee payment status.

2. Submission of Exam Application

- Filling out the exam form manually or online.

3. Payment of Examination Fees

- Paying via bank challan, online portal, or college counter.

4. Approval by Faculty/Exam Cell

- Faculty verifies student details and eligibility.

5. Generation of Admission Ticket

- Exam cell generates hall ticket with exam schedule and student details.

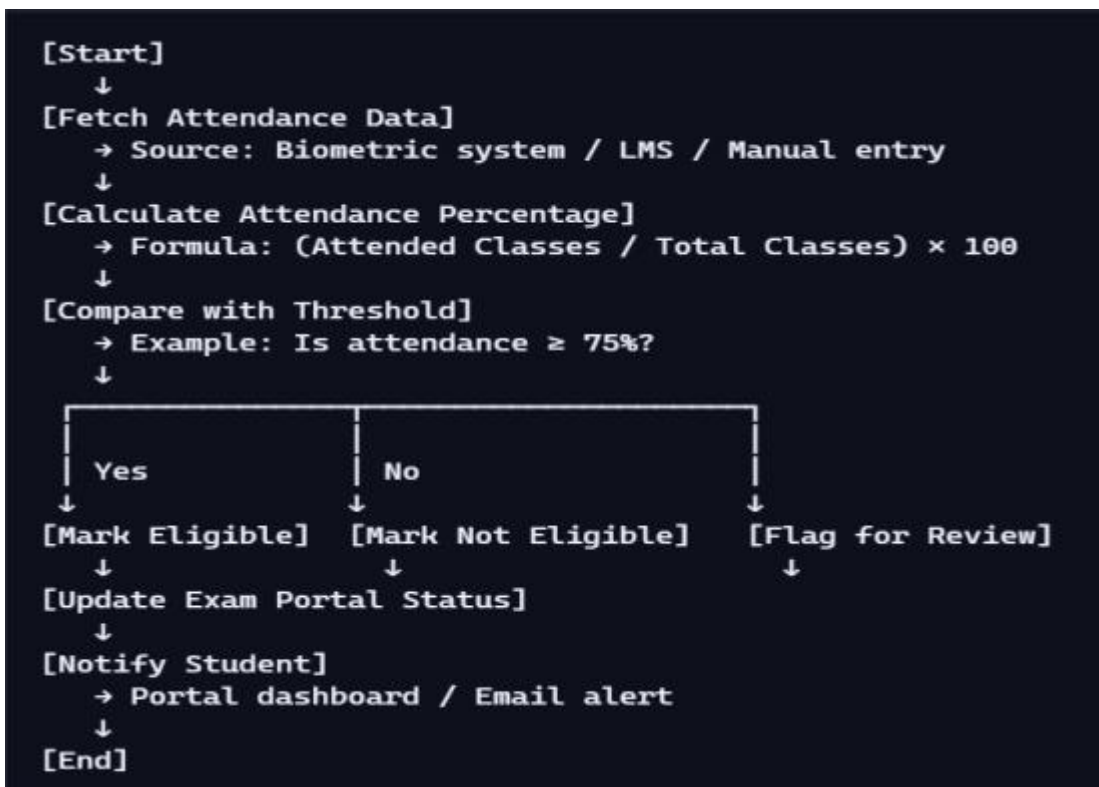
6. Distribution of Admission Ticket

- Tickets are printed and handed out physically or sent via email/portal.

Tasks That Can Be Automated:

Task	Sub-Tasks	Can Be Automated?	Automation Notes
1. Verification of Eligibility	Attendance, internal marks, fee status	✓ Yes	Auto-check via LMS, finance system, and attendance logs.
2. Submission of Exam Application	Online or manual form filling	✓ Yes	Online portal with validation and auto-save features.
3. Payment of Examination Fees	Bank challan, online payment, counter payment	✓ (partially) Yes	Online payment gateway integration; manual payment may still need verification.
4. Approval by Faculty/Exam Cell	Manual review of eligibility	× No	Requires academic judgment and manual verification.
5. Generation of Admission Ticket	Creating hall ticket with schedule and student details	✓ Yes	Auto-generate PDF based on verified data.
6. Distribution of Admission Ticket	Physical handout or digital delivery	✓ Yes	Email or portal-based download system.

Task 1: Illustration of Automation: Verification of Eligibility:-



Benefits of Automating Tasks:

- **Saves Time** – Faster processing and fewer delays.
- **Reduces Errors** – Accurate data handling and ticket generation.
- **Improves Access** – Students can download tickets anytime.
- **Enhances Efficiency** – Staff workload is reduced, improving overall system flow.

2b. Identify the following cloud service types and list their characteristics and advantages. Cisco WebEx Google App Engine, Amazon EC2.

➡ 1. Cisco WebEx — *Software as a Service (SaaS)*

Type: SaaS

Use Case: Collaboration and communication platform

Characteristics:

- Cloud-native architecture with microservices
- Offers video conferencing, messaging, webinars, and contact center solutions
- Integrated with Cisco's broader ecosystem (e.g., WebEx Calling, WebEx Contact Center)
- AI-powered features like real-time transcription, translation, and gesture recognition
- Available via browser, desktop, and mobile apps

Advantages:

- No infrastructure management required by users
- Scalable for small teams to large enterprises
- High security and compliance standards (e.g., end-to-end encryption)
- Seamless integration with other Cisco tools and third-party apps
- Ideal for hybrid and remote work environments

2. Google App Engine — *Platform as a Service (PaaS)*

Type: PaaS

Use Case: Deploying and scaling web applications

Characteristics:

- Fully managed serverless platform
- Supports multiple languages: Python, Java, Go, Node.js, PHP, Ruby
- Offers two environments: Standard (fast startup, auto-scaling) and Flexible (custom runtimes, Docker support)
- Built-in services: task queues, scheduled jobs, traffic splitting, and integration with Google Cloud services

Advantages:

- Automatic scaling based on traffic
- No server management—focus purely on code
- Tight integration with other GCP services like BigQuery, Cloud SQL, Firestore
- Pay-as-you-go pricing model
- High availability and reliability with Google’s infrastructure

3. Amazon EC2 — *Infrastructure as a Service (IaaS)*

Type: IaaS

Use Case: Provisioning virtual machines and compute resources

Characteristics:

- Offers a wide range of instance types (general-purpose, compute-optimized, memory-optimized, etc.)
- Supports custom AMIs, bare metal instances, and GPU acceleration
- Integrated with AWS services like S3, RDS, VPC, IAM
- Auto Scaling and Elastic Load Balancing for dynamic resource management
- Multiple pricing models: On-Demand, Reserved, Spot Instances

Advantages:

- Full control over the operating system and software stack
- Highly customizable for any workload—from web hosting to machine learning
- Global availability across AWS regions and availability zones
- Cost optimization through flexible pricing and resource provisioning
- Ideal for enterprises needing granular control over infrastructure